

ReSceneScribe: Static–Dynamic 3D Accident Reconstruction from Vehicle-Mounted Cameras and Multi-Agent Simulation for Evidence-Grounded Investigation

Woosup Lee^a, Yongmin Kim^{b,*}

^aPlanby Technologies Inc., 18 Yeoksam-ro 12-gil, Gangnam-gu, Seoul, Republic of Korea

^bDepartment of Civil and Environmental Engineering, KAIST, Daejeon, Republic of Korea

Abstract

Traffic accident investigation requires metric information on vehicle positions, road geometry, and collision relationships. However, accident scenes are often cleared rapidly, forcing investigators to rely on partial videos captured by vehicle-mounted cameras. This study presents ReSceneScribe, a static–dynamic 3D reconstruction framework that converts fragmented accident observations into two practical outputs: an interpretable temporal replay and a collision-state 3D asset for evidence-grounded reporting. The proposed framework combines two complementary data regimes. A pilot experiment using two real vehicles equipped with windshield-mounted cameras verifies that ordinary vehicle-mounted sensing can support static scene reconstruction under practical capture conditions. Since repeated physical collision data collection is unsafe and inherently limited, a four-vehicle scenario from DeepAccident Town03 is further used to evaluate dynamic accident reconstruction in a controlled multi-agent environment. The framework aligns multi-agent observations within a shared metric coordinate system, removes dynamic objects from the static road background, reconstructs vehicle geometry through vehicle-local motion compensation, and constrains report generation using frame-indexed geometric facts. In the selected scenario, the two involved vehicles overlap during source frames 54–56, and the final collision-state frame shows an oriented-bounding-box gap of 0.4479 m. The final hybrid Gaussian/RGB reconstruction contains 1,177,009 vertices, comprising 1,100,000 static background points and 77,009 motion-compensated vehicle points. These results demonstrate that ReSceneScribe can support accident-scene reconstruction by linking field-feasible sensing, simulation-based dynamic validation, and auditable geometric reporting within a unified analysis framework.

Keywords: traffic accident reconstruction, 3D reconstruction, vehicle-mounted camera, static–dynamic decomposition, multi-agent sensing, Gaussian-compatible point cloud, evidence-grounded reporting

1. Introduction

Accident analysis depends on the spatial relationship among vehicles, lanes, road markings, impact locations, and surrounding infrastructure. The most reliable evidence is therefore not only a video record but a metric three-dimensional state of the scene. In real accident response, however, this state is rarely preserved for long. Vehicles and debris are cleared to reduce secondary risk and to restore road capacity, while road-surface marks and vehicle positions may be altered before detailed measurement is possible. As a result, many cases are interpreted from partial photographs, vehicle-mounted camera videos, and witness statements. These sources are useful, but a single camera view is fundamentally limited by perspective distortion, occlusion, and uncertain metric scale.

Conventional 3D accident documentation has used terrestrial laser scanning, UAV photogrammetry, and related surveying tools to preserve scene geometry [1, 2]. These methods can produce accurate spatial records, but they require equipment, scene access, and operator time that are not always available in routine accident handling. Recent computer-vision methods provide a

complementary direction. Structure-from-Motion estimates camera motion and sparse geometry from image correspondences [3]; Neural Radiance Fields recover continuous radiance and density fields from posed images [4]; and 3D Gaussian Splatting represents scenes with explicit Gaussian primitives for efficient rendering [5]. Dynamic extensions such as 4D Gaussian Splatting further model time-varying scenes [6]. However, direct application of these methods to accident analysis remains insufficient because an accident scene is not merely a visual rendering target. It must preserve which vehicle moved where, which road geometry was static, when contact occurred, and which coordinates support each textual conclusion.

This paper therefore focuses on an application-oriented reconstruction problem: how to turn practical vehicle-mounted sensing and simulation-based accident data into an interpretable and reusable accident reconstruction output. The key design principle is static–dynamic separation. The road environment is reconstructed as a stable static asset after removing moving-object traces, while vehicles are represented separately by frame-wise poses and by motion-compensated collision-state geometry. This prevents the common failure mode in which moving vehicles are fused into the background and appear as repeated or blurred shapes.

The study uses two complementary data sources. First, a

*Corresponding author.

Email address: yongmin@kaist.ac.kr (Yongmin Kim)

physical pilot experiment with two real vehicles and windshield-mounted cameras is used to verify the practicality of vehicle-mounted acquisition for static road-scene reconstruction. This field component is important because it establishes that the sensing assumption is not purely synthetic. Second, a DeepAccident scenario generated in the CARLA simulator is used for the dynamic collision stage [7, 8]. DeepAccident is suitable here because it provides synchronized multi-view camera, LiDAR, calibration, and label data from multiple vehicles, enabling repeatable accident reconstruction without creating unsafe physical collisions.

The main contributions of this study are as follows: (1) a compact static–dynamic decomposition pipeline for converting multi-agent accident observations into a clean road background and dynamic vehicle trajectories; (2) a vehicle-local motion compensation strategy for reconstructing collision-state vehicle geometry while avoiding world-space ghosting artifacts; (3) a dual-output accident-analysis application that provides an interactive temporal replay for human interpretation and a Gaussian-compatible RGB PLY asset for reusable 3D inspection; and (4) an evidence-grounded reporting mechanism that constrains accident statements using vehicle identifiers, frame indices, metric coordinates, and collision diagnostics. Together, these contributions establish ReSceneScribe as an integrated accident reconstruction framework that links multi-view 3D reconstruction, dynamic collision-state modeling, and auditable report generation within a unified metric analysis workflow.

2. Method

2.1. Framework overview

The proposed framework is organized around the distinction between what remains fixed and what moves. Static elements include the road surface, lane markings, buildings, roadside structures, and fixed infrastructure. Dynamic elements include the target vehicles and other labelled moving objects. Treating all observations as one static point set would maximize the number of points but would destroy accident interpretability, because vehicle surfaces would be accumulated at several positions. ReSceneScribe therefore constructs two linked representations: a static background map and a dynamic vehicle representation.

The pipeline has four stages. First, vehicle-mounted camera or multi-agent sensor observations are converted into a common metric coordinate system. Second, static road geometry is reconstructed after removing points inside dynamic object boxes. Third, target vehicle geometry is accumulated in each vehicle’s local coordinate frame and then transformed to the selected collision pose. Fourth, geometric facts extracted from the reconstruction are used to constrain the accident report. Fig. 1 summarizes this workflow.

2.2. Metric alignment of multi-agent observations

For each vehicle agent a and frame t , calibration provides an ego-to-world transform $T_{a,t}^w$. A LiDAR point $p_{a,t}^\ell$ is mapped to the shared world coordinate system by

$$p_{a,t}^w = T_{a,t}^w T_{\ell,a,t}^e p_{a,t}^\ell, \quad (1)$$

where $T_{\ell,a,t}^e$ is the LiDAR-to-ego transform. Camera color is assigned by projecting the world-space point into synchronized camera images through the corresponding LiDAR-to-camera and intrinsic matrices. Points that are visible in a valid camera image receive RGB values from the projected pixel, while points without valid projection use an intensity-based fallback. The output of this stage is not only a point cloud; it is a frame-indexed evidence record containing 3D coordinates, source agent, source frame, color, and semantic or object labels where available.

In the physical pilot, the same principle is applied at a smaller scale using vehicle-mounted camera reconstruction and scene alignment. The pilot does not attempt to create collision dynamics; its role is to validate that field-acquired vehicle-mounted imagery can support road-scene reconstruction and camera-pose reasoning. In the simulation experiment, the calibration is known and enables direct multi-agent fusion in the DeepAccident world coordinate system.

2.3. Static background reconstruction

The static background is generated by fusing points from all available agents and frames after removing dynamic objects. Let $\mathcal{P}_{a,t}^w$ be the set of world-space points at frame t from agent a , and let $\mathcal{B}_{d,t}$ be the set of 3D bounding boxes corresponding to dynamic objects. A point is accepted into the static set only if it is outside all dynamic boxes:

$$\mathcal{S} = \{p \in \mathcal{P}_{a,t}^w : p \notin B, \forall B \in \mathcal{B}_{d,t}\}. \quad (2)$$

This filtering also removes points inside the target vehicle boxes, including self-vehicle observations, because target vehicles are reconstructed separately. The filtered points are then voxel-downsampled to control density while preserving road topology and large-scale structural features.

This stage is central to the contribution. It converts multi-agent accident sensing into a stable background representation that can be inspected independently of the vehicles. It also prevents a misleading reconstruction in which dynamic vehicles appear as multiple semi-transparent traces embedded in the road environment.

2.4. Vehicle-local motion compensation

Target vehicle geometry is reconstructed using the opposite strategy. Instead of discarding points inside target vehicle boxes, these points are collected as vehicle observations. For a target vehicle k , any point $p_{k,t}^w$ inside its oriented bounding box at frame t is transformed into the vehicle-local coordinate system:

$$p_{k,t}^v = (T_{k,t}^w)^{-1} p_{k,t}^w, \quad (3)$$

where $T_{k,t}^w$ denotes the pose of vehicle k . Points are filtered to reduce road-surface contamination under the vehicle body, accumulated across frames and observing agents, and voxel-downsampled in local coordinates. The downsampled local geometry $\tilde{p}_k^v$ is then transformed to the selected collision-state pose T_{k,t_c}^w :

$$\hat{p}_k^w = T_{k,t_c}^w \tilde{p}_k^v. \quad (4)$$

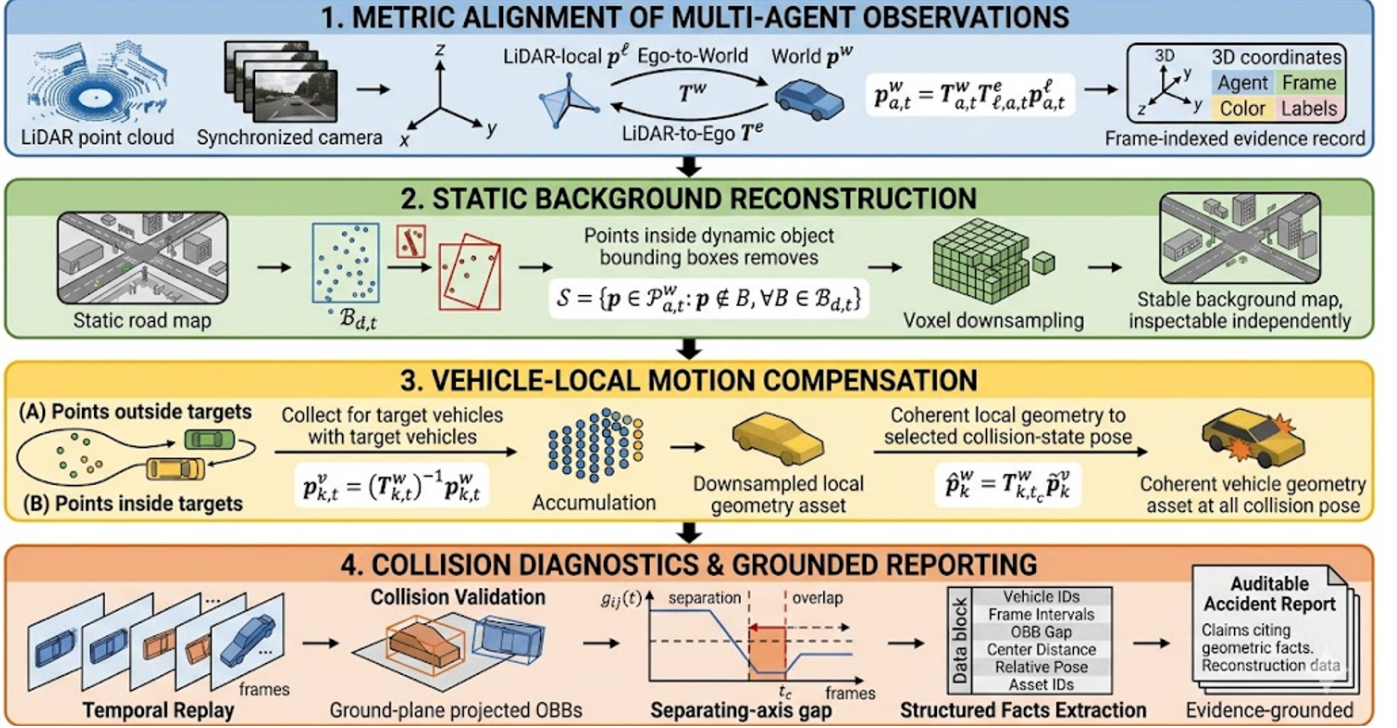


Figure 1: Overview of ReSceneScribe. A physical pilot verifies practical vehicle-mounted acquisition, while the simulation scenario supplies repeatable dynamic accident observations. The method separates static road geometry from dynamic vehicle motion and exports both an interpretable temporal replay and a collision-state 3D asset for evidence-grounded accident reporting.

This procedure is motion compensation in the accident-reconstruction sense. If vehicle points were accumulated directly in the world frame, the resulting geometry would contain a trail along the vehicle trajectory. By accumulating in the vehicle-local frame, each observed vehicle surface contributes to one coherent vehicle geometry, which is then placed at the final collision pose. The method therefore preserves both temporal motion for replay and spatial geometry for final accident-state inspection.

2.5. Collision diagnostics and grounded reporting

The temporal replay uses frame-wise vehicle poses to animate all participating vehicles. Collision validation is performed using oriented bounding boxes (OBBs) projected to the ground plane. Let $g_{ij}(t)$ be the minimum separating-axis gap between vehicles i and j at frame t . A positive value indicates separation, while a negative value indicates overlap. The selected collision-state frame is the last available frame in which the collision pair is in a validated overlap state.

The reporting layer is designed to avoid unrestricted visual narration. The system first extracts structured facts such as vehicle identifiers, frame indices, OBB gap, center distance, relative pose, and reconstructed asset identifiers. These facts are serialized into a report context, and natural-language descriptions are constrained to cite them. A statement such as a vehicle entering an occupied conflict area is therefore associated with a vehicle ID, a frame interval, and a geometric measurement. The purpose is not to replace legal judgement, but to support an auditable

accident report whose claims are grounded in reconstructed geometry.

3. Experiments and Results

3.1. Physical pilot experiment

The physical pilot experiment was conducted to examine the feasibility of vehicle-mounted image acquisition in a real road environment before relying on a simulated collision dataset. Two real vehicles, a Hyundai Avante and a Hyundai Casper, were used. Smartphone cameras were mounted near the windshield area to approximate ordinary vehicle-mounted camera placement. The cameras recorded RGB videos at a resolution of 1920×1080 pixels and 30 frames/s. The vehicles performed controlled approach and avoidance maneuvers in a non-collision setting.

This pilot experiment has a specific role in the paper. It does not serve as the quantitative collision benchmark because physical impact data are difficult and unsafe to collect repeatedly. Instead, it verifies three practical assumptions: vehicle-mounted images can capture useful road-scene structure, the reconstruction workflow can operate without specialized forensic survey equipment, and static road geometry can be conceptually separated from moving vehicle behavior. These observations define the practical sensing side of the proposed application, while the simulation experiment supplies the dynamic accident reconstruction side.



(a)



(b)

Figure 2: Physical pilot experiment and static scene reconstruction using vehicle-mounted RGB cameras. (a) Real-world pilot setup with two vehicles, a Hyundai Casper and a Hyundai Avante, used to acquire vehicle-mounted visual observations under controlled approach and avoidance conditions. (b) Static 3D reconstruction result generated from the vehicle-mounted RGB video, showing the recovered road environment and the relative configuration of the two vehicles.

Table 1: Essential experimental configuration.

Item	Configuration
Physical pilot	Two real vehicles; windshield-mounted smartphone cameras; 1920×1080; 30 frames/s; controlled non-collision approach and avoidance maneuvers
Simulation scenario	DeepAccident Town03, type001/subtype0001/scenario00024
Scene condition	Four-way junction, MidRainSunset
Simulation agents	Four vehicle agents: two collision vehicles and two following vehicles
Available source frames	001–056
Main outputs	Interactive temporal replay; collision-state hybrid Gaussian/RGB PLY

3.2. DeepAccident simulation scenario

The simulation experiment uses the DeepAccident mini/local scenario Town03_type001_subtype0001_scenario00024. The scenario is a four-way junction under MidRainSunset conditions. It contains four vehicle agents: two collision vehicles and two following vehicles. Each vehicle provides 56 available frames with multi-view RGB cameras, LiDAR, calibration files, and 3D object labels. The source frame range used in this study is 001–056.

The official metadata accident frame is beyond the available mini subset. Therefore, the collision state is determined geometrically from the available frames rather than assumed from metadata. OBB analysis shows overlap between the two collision vehicles during source frames 54–56. Frame 56 is selected as the final collision-state frame because it is the last available frame with validated overlap.

Table 2: Vehicle-level motion and reconstructed geometry summary.

Vehicle role	Net motion (m)	Final points
Collision vehicle A	39.9009	28,979
Follower behind A	32.6585	8,024
Collision vehicle B	43.8988	28,889
Follower behind B	41.2916	11,117

Table 3: Core quantitative results used for accident interpretation.

Metric	Value
Verified OBB overlap frames	54–56
Selected collision-state frame	56
Final OBB gap	−0.4479 m
Final center distance	4.2299 m
Source LiDAR points	8,616,764
Removed dynamic/target-vehicle points	312,543
Final PLY vertices	1,177,009
Static / vehicle PLY points	1,100,000 / 77,009

3.3. Temporal accident replay

The first output is an interactive four-vehicle accident replay. The static background is reconstructed by fusing LiDAR streams from all four vehicle agents after dynamic-object removal. Vehicle motion is replayed using calibration-derived frame-wise poses and dimension-aware vehicle proxies. This output is intended for human interpretation: a user can inspect the road layout, vehicle trajectories, and synchronized vehicle-mounted camera frames in one view.

The four vehicle trajectories also verify that the scenario is not a single-view reconstruction problem. The two collision vehicles approach the intersection from different directions, while the following vehicles provide additional viewpoints and context. In the selected frame range, the collision vehicles overlap from frame 54 to frame 56. At frame 56, the final OBB gap is −0.4479 m and the center distance is 4.2299 m, supporting the use of frame 56 as the final collision-state frame.

3.4. Collision-state hybrid Gaussian/RGB reconstruction

The second output is a collision-state 3D asset. The method starts from 8,616,764 source LiDAR points collected across the four vehicle agents. During static background generation, 312,543 dynamic or target-vehicle points are removed to suppress moving-object contamination. The final PLY contains 1,177,009 vertices: 1,100,000 static background points and 77,009 vehicle points reconstructed through vehicle-local motion compensation. The PLY includes ordinary RGB fields for general point-cloud viewers and Gaussian-compatible fields such as DC color, opacity, scale, and rotation for downstream Gaussian-style rendering workflows.

Vehicle point density differs by vehicle because LiDAR visibility, occlusion, and viewing distance differ. This is expected in an accident scene and should not be hidden by over-smoothing. Instead, the final asset preserves the actually observed geometry while placing each vehicle at the validated collision pose. Table 2 summarizes the vehicle-level reconstruction counts.



Figure 3: Simulation-based accident replay from the DeepAccident Town03 four-vehicle collision scenario. The visualization combines a fused static LiDAR background, frame-indexed vehicle trajectories, collision-state bounding boxes with an overlap of approximately -0.45 m, and synchronized vehicle-mounted RGB observations from the four agents.

3.5. Application-oriented interpretation

The important result is not only the number of reconstructed points. The contribution is that the accident is represented in a form that supports analysis. The replay explains when and how the four vehicles approached the conflict area. The collision-state PLY preserves the static road context and the observed vehicle geometries at the final overlap frame. The collision diagnostics provide frame-indexed evidence that can be cited in a report. Therefore, the system converts multi-agent accident data into a practical investigation asset rather than a dataset-specific visualization.

4. Discussion

The proposed framework is intentionally different from a general-purpose visual reconstruction pipeline. General neural rendering methods often prioritize photorealistic novel-view synthesis. Accident reconstruction additionally requires metric traceability. A visually plausible reconstruction is not enough if the report cannot identify which frame, coordinate, or vehicle measurement supports a claim. By maintaining a static background, dynamic vehicle trajectories, collision diagnostics, and structured reporting facts, ReSceneScribe keeps the visual, geometric, and linguistic components tied to the same evidence base.

The use of both physical and simulation data is also a methodological contribution. A purely physical accident dataset would

be difficult to collect safely and repeatedly. A purely simulated study, however, could be criticized as detached from practical sensing assumptions. The field pilot addresses the acquisition feasibility of ordinary vehicle-mounted cameras, while DeepAccident provides repeatable dynamic collision data with multi-agent calibration. The combined design therefore reflects a realistic development path for accident reconstruction technology: test sensing in the field, validate collision logic in simulation, and connect both to interpretable reporting outputs.

Several limitations remain. First, the final PLY is a deterministic hybrid Gaussian/RGB representation rather than a fully optimized dynamic 3D Gaussian Splatting model. It is designed for metric stability and viewer compatibility, not for maximum photorealism. Second, vehicle surface completeness depends on LiDAR visibility; following vehicles or heavily occluded surfaces may have lower point density. Third, collision validation uses OBB overlap as a geometric proxy. This is appropriate for scenario-level accident-state verification, but future deployment should incorporate deformation, contact surface estimation, and uncertainty propagation. Fourth, the reporting module depends on the accuracy of extracted geometric facts and should not be interpreted as an automated legal fault decision system.

5. Conclusion

This paper presented ReSceneScribe, a static–dynamic 3D reconstruction framework for multi-vehicle accident replay and evidence-grounded reporting. The work contributes

an application-oriented pipeline that connects a real-world vehicle-mounted camera pilot with simulation-based dynamic collision reconstruction. The physical pilot demonstrates the feasibility of practical road-scene capture using ordinary vehicle-mounted cameras, while the DeepAccident Town03 scenario supplies controlled four-vehicle collision dynamics. The method separates static road geometry from dynamic vehicles, reconstructs vehicles using vehicle-local motion compensation, and exports both an interactive replay and a collision-state hybrid Gaussian/RGB PLY.

The main technical contribution is the reconstruction structure: static background fusion is protected from moving-object ghosting, and vehicle geometry is accumulated in local coordinates before placement at the final collision pose. The main application contribution is the evidence structure: accident descriptions can be tied to vehicle identifiers, frame indices, OBB overlap, and metric coordinates. In the selected scenario, the final asset contains 1,177,009 vertices and the collision pair is validated by a final OBB gap of -0.4479 m at frame 56.

The current limitations are clear. The framework does not yet provide a fully time-continuous dynamic Gaussian model, vehicle surface density is limited by sensor visibility, and OBB overlap does not model physical deformation or impact force. Future work will extend the hybrid PLY into time-varying Gaussian primitives, add uncertainty-aware collision metrics, evaluate reconstructed vehicle geometry against label boxes using Chamfer distance or IoU, and integrate vehicle telemetry or V2X messages when available. These extensions will move the system from accident visualization toward a more complete forensic decision-support tool.

Code availability

The implementation of ReSceneScribe will be made available at: <https://github.com/Kimyongmin1018/IEEE-IES-Generative-AI-challenge-ReSceneScribe.git>.

Declaration of generative AI and AI-assisted technologies

Generative AI assistance was used only to support the visual drafting and layout refinement of Figure 1. The authors reviewed and finalized the figure and take full responsibility for the scientific content of this article.

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